PHYSICS 2111 · PHYSICS NINJA PRACTICE EXAM

Exam 2 · Version B · Solutions

20 questions · 90 minutes · original practice exam, not an official exam

ANSWER KEY

1	**a**	2	**c**	3	**e**	4	**c**	5	**c**	6	**b**	7	**b**
8	**e**	9	**c**	10	**d**	11	**c**	12	**b**	13	**d**	14	**e**
15	**b**	16	**b**	17	**a**	18	**b**	19	**a**	20	**a**		

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 4 kg block is released from rest at the top of a ramp 3 m high. On the way down, friction does −40 J of work on it.

1) What is the speed of the block at the bottom of the ramp?

Answer: (a) 6.23 m/s

$\frac{1}{2}mv^2 = mgh - 40 = 77.7$ J, $v = 6.23$ m/s.

2) The change in the mechanical energy of the block is

Answer: (c) −40 J

Non-conservative work changes mechanical energy: $\Delta E = W_{friction} = -40$ J.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 2 kg block slides at 3 m/s across a frictionless floor and runs into a spring with k = 450 N/m.

3) What is the maximum compression of the spring?

Answer: (e) 0.2 m

$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$, $x = v\sqrt{(m/k)} = 0.2$ m.

4) What is the speed of the block when the spring is compressed 0.10 m?

Answer: (c) 2.6 m/s

$\frac{1}{2}mv^2 = 9 - \frac{1}{2}(450)(0.1^2) = 6.75$ J, $v = 2.6$ m/s.

5) The magnitude of the block's acceleration is largest

Answer: (c) at maximum compression.

The spring force kx is largest when x is largest.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A force acts on a 3.0 kg cart along the x axis. It rises linearly from 0 at x = 0 to 30 N at x = 4.0 m, then stays at 30 N until x = 6.0 m. The cart starts from rest on a frictionless track.

6) How much work does the force do from x = 0 to x = 6.0 m?

Answer: (b) 120 J

Area under F(x): ½(4)(30) + (2)(30) = 60 + 60 = 120 J.

7) What is the speed of the cart at x = 6.0 m?

Answer: (b) 8.94 m/s

½(3)v^2 = 120 J, v = 8.94 m/s.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 60 kg skier leaves the top of a 40 m high frictionless hill moving at 5.0 m/s.

8) What is the skier's speed at the bottom?

Answer: (e) 28.5 m/s

$v = \sqrt{(v_0^2 + 2gh)} = \sqrt{(25 + 784.8)} = 28.5$ m/s.

9) A heavier skier starting the same way would reach the bottom with a speed that is

Answer: (c) the same.

The mass cancels in the energy equation.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

An 80 kg astronaut floating at rest outside a space station throws a 4.0 kg wrench at 6.0 m/s.

10) What is the astronaut's recoil speed?

Answer: (d) 0.3 m/s

Momentum: 80v = (4)(6), v = 0.3 m/s.

11) After the throw, the kinetic energy of the wrench is

Answer: (c) greater than the astronaut's.

Equal momenta: K = p^2/2m, so the smaller mass has more kinetic energy.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 1000 kg car moving east at 15 m/s collides at an intersection with a 1500 kg car moving north at 10 m/s. The cars stick together.

12) What is the speed of the wreck just after the collision?

Answer: (b) 8.49 m/s

p_x = 15 000, p_Y = 15 000 kg·m/s. $v = \sqrt{(p_x^2 + p_Y^2)}/2500 = 8.49$ m/s.

13) In what direction does the wreck move just after the collision?

Answer: (d) 45.0° north of east

$p_Y = p_x$, so the angle is 45° north of east.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

Three small objects lie on the x axis: 2.0 kg at x = 0, 3.0 kg at x = 2.0 m and 5.0 kg at x = 4.0 m.

14) Where is the center of mass?

Answer: (e) 2.6 m

x_cm = (2·0 + 3·2 + 5·4)/10 = 2.6 m.

15) The objects sit on frictionless ice and push each other apart. The center of mass

Answer: (b) stays where it is.

Internal forces cannot move the center of mass of an isolated system.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 0.020 kg bullet is fired into a 2.0 kg wooden block hanging from a long string. The bullet embeds in the block, and the block swings up 0.12 m.

16) What was the bullet's speed just before impact?

Answer: (b) 155 m/s

After: v = √(2gh) = 1.53 m/s. Before: v_bullet = (2.02/0.02)(1.53) = 155 m/s.

17) During the very short collision itself, which quantity is conserved?

Answer: (a) Momentum only

The collision is perfectly inelastic: momentum is conserved, kinetic energy is not.

THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.

A car moves at 72 km/h. Its tires have a radius of 0.35 m and roll without slipping.

18) What is the angular speed of the tires?

Answer: (b) 57.1 rad/s

v = 20 m/s; ω = v/R = 57.1 rad/s.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A solid disk of mass 8.0 kg and radius 0.30 m spins at 120 rpm about its central axis.

19) What is its rotational kinetic energy?

Answer: (a) 28.4 J

$I = \frac{1}{2}MR^2 = 0.36\ kg\cdot m^2$, ω = 12.6 rad/s, $K = \frac{1}{2}I\omega^2 = 28.4$ J.

20) A thin ring with the same mass and radius spinning at the same rate has a kinetic energy that is

Answer: (a) twice as large.

$I_{ring} = MR^2$ is twice $I_{disk} = \frac{1}{2}MR^2$.
